05: Conditions - live

Stat 230 - Spring 2026

> **Note**
>
> This is the tutorial version of the R activity. If you prefer to use .qmd and RStudio directly, you can use https://aluby.github.io/stat230/activities/05-conditions

```
library(ggformula)
theme_set(theme_minimal())
```

1 Example 1: Making residual plots in R

Recall the `movies` dataset from the R activity we did on Day 02. (head() prints out the first few rows)

```
movies <- read.csv("https://aloy.github.io/stat230-materials/data/movie_scores.csv")
head(movies)
```

On that day, we made the following scatterplot:

```
gf_point(audience ~ critics, data = movies) |>
  gf_lm()
```

and fit this regression model:

```
movie_lm <- lm(audience ~ critics, data = movies)
movie_lm
```

1.1 Finding residuals

In order to make residual plots, we first need to find the residuals. While we can do this by hand for a couple of points, it would be extremely tedious to do it for *every single data point*! Instead, we'll use the `augment()` function from the {broom} R package to created an augmented movie dataset. This takes the original X and Y variables and adds the $\hat{y}$ values (`.fitted`) and residuals (`.resid`) (along with some other columns that we'll use later)

Run the following code chunk and take a peek at the `movie_aug` dataset. Can you find the:

- original X variable?
- original Y variable?
- Predictions?
- Residuals?

```
library(broom)
movie_aug <- augment(movie_lm)
movie_aug
```

1.2 Making residual plots

The code below makes a normal QQ plot and a histogram of the residuals. Do you think the normality assumption is satisfied?

```
#| layout-ncol: 2


gf_qq(~.resid, data = movie_aug) |>
  gf_qqline()

gf_histogram(~.resid, data = movie_aug, col = "white")
```

1.2.1 Checkpoint question

Use the `movie_aug` dataset to make a scatterplot with `.fitted` on the x-axis and `.resid` on the y-axis. List the SLR conditions you can assess with this plot, and whether or not they appear to be satisfied.

```
# your code here
```

1.3 A shortcut

We will make a **ton** of residual plots this term. A shortcut is to use the `resid_panel` function within the {ggResidpanel} package:

```
library(ggResidpanel)
resid_panel(movie_lm)
```

1.3.1 Checkpoint question

Which panel(s) would you look at for each of the following SLR conditions. List as many panels as possible, along with how you would identify any violations:

- Linearity:
- Independence:
- Normality:
- Equal variance:

2 Example 2: R^2 in R

2.0.1 Checkpoint question

Find the R^2 value for the `movie_lm` model using `summary()`. Confirm that it is equal to r^2

```
# your code here
```

2.1 Summary

2.1.1 Checkpoint question

Suppose that we fit a linear regression model to a sample of 1,000 observations and we find evidence in the normal Q-Q plot of a violation of the normality assumption, but no evidence in the residual plot of violations of the linearity and constant variance assumptions and no reason to suspect a violation of the independence assumption. Which of the following inferences are valid, and which are not?:

- p-value from the hypothesis test $H_0 : \beta_1 = 0$ vs $H_A : \beta_1 \neq 0$
- Confidence interval for β_1
- Confidence interval for the mean of Y at some X value
- Prediction interval for Y at some X value

2.2 Function quick reference

The following table summarizes the functions we learned today:

Function	Purpose
`augment(model)`	Create augmented dataset that includes predictions and residuals
`gf_qq(~variable, data)`	Make a qq plot
`gf_qq_line()`	Overlay a straight line on the qq plot
`resid_panel(model)`	Shortcut for making a panel of residual plots